TP-Link Point-to-Multipoint Network Design

Long-Range 3 KM Wireless Deployment Blueprint for Remote Security Cameras

System Architecture & Requirements

This document provides a highly structured engineering layout and corresponding Bills of Materials (BOM) for establishing a high-throughput Point-to-Multipoint (PtMP) Wi-Fi network over a 3 KM radius. The primary objective is to link multiple remote security gate zones back to a central base station using dedicated TP-Link long-range hardware.

- **Clear Line of Sight (LoS):** At 3 KM, structural obstacles, dense foliage, or terrain contours will completely degrade performance. Base stations and receiving zones must have 100% clean visual paths.
- **Power Infrastructure:** All hardware operates via Power over Ethernet (PoE), allowing clean deployment inside outdoor enclosures powered by AC grid connections or isolated off-grid solar kits.
- **Scale Configuration:** Remote receiving zones require one client antenna per physical zone. Replace variables marked with 'N' in the BOMs below with your specific destination count.

Option 1: The Omada Enterprise Ecosystem (Recommended)

Designed for enterprise networks, providing unified management, robust multi-camera bandwidth, and high-density reliability over up to 5 KM paths. The remote receiver eliminates the need for field switches due to integrated triple multi-gigabit ports.

Part Number	Description / Hardware Specifications	Qty	Role & Engineering Implementation
Sector Bridge 5 Omada	Integrated base station + 120-degree sector antenna, 5GHz, up to 5 KM range, 867 Mbps capacity, TDMA traffic management.	1	Central Base Station: Handles up to 32 remote receiving clients simultaneously across a broad wide-angle plane.
Omada EAP215-Bridge	Remote client bridge antenna, 35-degree narrow directional beamwidth, 3x Gigabit Ethernet ports.	N	Remote Receiving Zones: Installed at each camera destination point. Tri-ports handle direct camera connection without local field switches.

Option 2: The Pharos Standalone Architecture

A standalone option using individual local interfaces without needing a unified controller. Utilizes high-gain physical dishes and clip-on sector fields to deliver massive noise isolation across open layouts.

Part Number	Description / Hardware Specifications	Qty	Role & Engineering Implementation
Pharos WBS510	Base station radio unit, 5GHz dual-polarized external connectors, IP65 weatherproofing.	1	Central Base Station Core: Mounted directly to the sector antenna to drive high-power signals down long property lines.
TL-ANT5819MS	19dBi 5GHz outdoor sector antenna, 120-degree horizontal beamwidth, toolless clip-on assembly.	1	Central Sector Array: Connects to WBS510 base station to shape the wide coverage field for receiving zones.
Pharos CPE710	High-gain 23dBi directional dish reflector antenna, 5GHz, 867 Mbps max data throughput, 802.11ac protocols.	N	High-Bandwidth Remote Node: Standard choice for heavy-bitrate multipoint applications like multi-lens 4K configurations.
Pharos CPE510 (Alt)	Compact 13dBi internal dual-polarized directional panel antenna, 5GHz, 300 Mbps capability.	N	Budget Remote Node Alternative: Deployed instead of CPE710 at light data destinations running single or low-bitrate streams.

Essential Deployment Accessories Checklist

A proper long-range installation requires outdoor-grade cabling and surge protection primitives to ensure survival against lightning strikes and element exposure. Ensure the deployment pipeline accounts for:

- **Outdoor Shielded Cat6 Cable (FTP/STP):** Mandatory for connecting units down structural poles to minimize cross-talk and survive UV degradation.
- **In-Line RJ45 Ethernet Surge Protectors:** Placed immediately before cables breach local equipment boxes to isolate dangerous ground potentials during electrical storms.
- **PoE Injectors:** Included in basic product retail boxes; ensure they are firmly housed inside NEMA-rated weatherproof enclosures at remote locations.